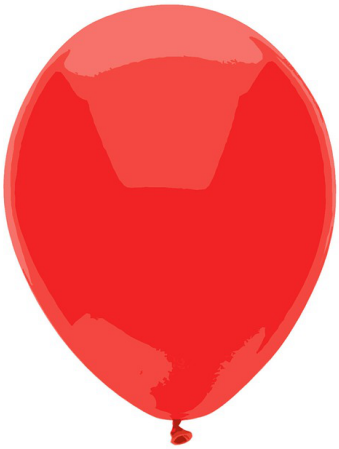
5
Gases



THE air we all breath contains numerous gases, such as oxygen, nitrogen or carbon dioxide. Some of these gases are indeed essential for life. As an example, plants take up carbon dioxide to give off oxygen, and water is produced by the reaction of oxygen and hydrogen gas. Other gases are dangerous for life. An example is carbon monoxide, which results from gas stoves, heating systems and fire. This is a colorless, odorless, and tasteless gas that can bind to the blood displacing oxygen. As a consequence, carbon monoxide can build up in closed environments causing death. This chapter deals with the properties of gases. You will learn how to calculate the volume or pressure of a gas, characterizing its state. You will also learn how to work with mixtures of gases and for example predict the pressure of oxygen in an atmosphere containing numerous gases.

GOALS

Chemistry and molecules

1. Apply the ideal gas law in gas calculations

Acids and bases

1. Predict the impact of CO_2 on acidity

Electrolytes and solutions

1. Calculate partial pressures

Reactions and energy

1. Balance chemical reactions

The chemistry of cancer

1. Learn about the dangers of radon

Let's calculate

1. Convert square and cubic units
2. Use conversion factors involving volume units

5.1 Gases and its properties

Gases contain atomic or molecular particles. They have very different properties than liquids or solids. The particles of a gas are spread and far away from each other. Liquids, on the other hand are made of loose particles that interact by means of weak forces. Solids on the other hand are packed materials and its particles, atoms or molecules, are closer together. This section covers the different properties of gases.

`Gases in the periodic table` Some of the elements in the periodic table are molecular gases, resulting of the combination of two atoms of the same element. For example, molecular oxygen (O_2) is gas. Similarly, molecular nitrogen (N_2), molecular hydrogen (H_2), molecular chlorine (Cl_2), or molecular fluorine (F_2) are all diatomic gases–they contain two atoms of the same element. Other gases result of the combination of two different non-metals. Examples are carbon monoxide (CO) or dioxide (CO_2), and nitrogen monoxide (NO) or dioxide (NO_2). The nobel gases (Ne, He, Ar) also exist in gas state.

`Characteristics of gases` Gases has different properties compared to solids of liquids:

- Gases assume the volume and shape of its container. As they expand, they have no shape different than their container's shape.
- Gases are compressible: they can be compressed, reducing its volume. Differently, liquids and solids are incompressible.
- The density of gases is small, compared to the one for solids and liquids.

▼Chlorine reactive is a gas

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▼Hydrogen is a flammable gas used to power some cars

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`Volume, V` The volume of a gas (V) is the amount of space it occupies, and gases fully occupy the volume of its container. Liters (L) and milliliters (mL) are units of volume. Liter is a cubic unit and one litter equals to a cubic decimeter (dm^3).

`Temperature, T` The temperature (T) of a gas is related to the speed (the average velocity) of its particles. The higher the temperature the higher the particles' speed. Although there are different units of temperature such as Kelvin (K) or celsius (C°), in this chapter many formulas require the use of Kelvin temperature (T_K), that related to celsius (T_C) by the formula

$$\boxed{T_K = T_C + 273}$$

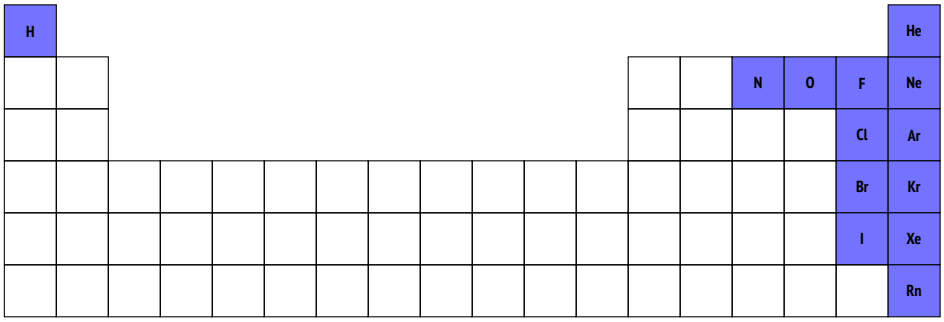

Figure 5.1 All gas elements

`Amount of gas, n` The amount of a gas (n) refers to the quantity of gas particles. The larger the amount of gas, the larger the amount of gas particles. The amount of gas is measured in moles or grams. Figure 5.1 displays a list of all gas elements.

`Pressure, P` In general pressure is defined as force divided by surface. In the international system the unit of force is the Newton (N) and the unit of area (A) is the m^2. One newton is $1\text{kg} \cdot \text{m/s}^2$.

$$\boxed{P = \frac{F}{A}}$$

The particles of a gas are constantly moving. On its movement, they frequently hit the walls of its container, like the raindrops hitting the ceiling when it rains. When they hit the walls they exert a pressure, and pressure is defined as the force acting on certain area. The larger pressure the stronger the collisions with the walls and the higher the frequency of collision–the stronger the force applied on the walls. Imagine you are driving a motorcycle. While you drive you can feel the collision of the air's particle with your face. The faster you go the higher pressure. The value of air's pressure is measured with a barometer and depends on your location on the earth–in particular your altitude–as well as the weather. If you are at the sea level the atmospheric pressure is one unit of pressure (one atm), due to the air that you have on top of you. If you climb a mountain, the pressure decreases, as there is less air on top of you. The higher you are with respect to the sea level, the lower the air pressure. The weather also affects pressure, and in hot days the pressure of air is higher, whereas on a cold day pressure is lower.

Units of pressure are: bars, atmospheres (atm), torr, pascals (Pa) or millimeters of mercury (mmHg). In order to covert pressure units, you can use the following conversion factors:

$$\boxed{\frac{1\text{ atm}}{1.01325\text{ bar}}} \quad \boxed{\frac{1\text{ atm}}{760\text{ mmHg}}} \quad \boxed{\frac{1\text{ torr}}{1\text{ mmHg}}} \quad \boxed{\frac{1\text{ atm}}{101325\text{ Pa}}}$$

one millimeters of mercury (mmHg) is the same as 1 torr. As a note, the name torr knowledges the person who invented the barometer: Torricelli, an Italian physicist.

Sample Problem 1

An oxygen sample has a pressure of 2 atm. Convert this value to: (a) mmHg and (b) Pascals.

SOLUTION

(a) we start by placing the given data (2 atm) and using the conversion factor between atm and mmHg, with the atm unit on the bottom, so that the units cancel

$$2\ \cancel{\text{atm}} \times \frac{760\ \text{mmHg}}{1\ \cancel{\text{atm}}} = 1520\text{mmHg}$$

(b) we proceed as in (a) but using the conversion between atm and Pa:

$$2\ \cancel{\text{atm}} \times \frac{101325\ \text{Pa}}{1\ \cancel{\text{atm}}} = 2.02 \times 10^5\text{Pa}$$

STUDY CHECK

An oxygen sample has a pressure of 730 mmHg. Convert this value to atmospheres.

▼A barometer used to measure the atmospheric pressure

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▼A manometer used to measure gas pressures

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▼A pressure gauge

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Figure 5.2 Pressure measuring devices

5.2 Ideal gas law

Ideal gases are gases made of particles without a size (very tinny) that do not interact with each other. The temperature, pressure, volume and number of moles of a gas are not independent. They are related by the ideal gas law. In this section we will introduce this law in two different forms: in terms of volume and in terms of density.

Ideal gas law in terms of moles The ideal gas law says:

$$\boxed{PV = nRT} \quad \text{Ideal Gas Law}$$

where:

P is the pressure of the gas in atm

V is the volume of the gas in L

n is the number of moles of the gas

T is the temperature of the gas in K

R is the constant of the gas $0.082 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}}$

Imagine for example that you inflate a balloon with you mouth, introducing air particles into the balloon. While the number of particles inside the balloon grows, its volume will grow too. More particles will collide with the walls of the balloon and hence, the pressure inside the balloon will also increase.

Sample Problem 2

Helium gas is used to inflate blimps, scientific balloons and party balloons. What is the volume in litters of a 0.2 moles of Helium balloon at 300K and 2 atm.

SOLUTION

	Given	Asking
Analyze the Problem	$T = 300K$ $P = 2atm$ $n = 0.2mol$ $R = 0.082 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}}$	V

Using now the ideal gases formula: $PV = nRT$, we have

$$2\cancel{\text{atm}} \cdot V = 0.2\cancel{\text{mol}} \cdot 0.082 \frac{\cancel{\text{atm}} \cdot \text{L}}{\text{mol} \cdot \cancel{K}} \cdot 300\cancel{K}$$

All units but L cancel out. Solving for V we have 2.46 L.

STUDY CHECK

What is the pressure in atmospheres of a 1 L balloon containing 3 moles of Helium at 40C°.

`Ideal gas law in terms of density` The ideal gas law in terms of density is:

$$\boxed{P \cdot MW = DRT} \quad \text{Ideal Gas Law in terms of D}$$

where:

P is the pressure of the gas in atm

MW is the molecular weight (or atomic weight, AW) of the gas in g/mol

D is the density in $g \cdot L^{-1}$

T is the temperature of the gas in K

R is the constant of the gas $0.082 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}}$

We use this formula when we are questioned about the molar mass or density of the gas.

Sample Problem 3

What is the density of Helium balloon at 400K and 3 atm.

SOLUTION

Besides the data in the problem, as the gas is He we already know its atomic mass from the periodic table:

	Given	Asking
Analyze the Problem	$T = 400K$ $P = 3atm$ $AW = 4g \cdot mol^{-1}$ $R = 0.082 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}}$	D

Using now the ideal gases formula in terms of density: $P \cdot MW = DRT$, we have

$$3\cancel{\text{atm}} \cdot 4 \frac{g}{\cancel{\text{mol}}} = D \cdot 0.082 \frac{\cancel{\text{atm}} \cdot \text{L}}{\cancel{\text{mol}} \cdot \cancel{K}} \cdot 400\cancel{K}$$

Solving for D we have 0.36 $g \cdot L^{-1}$.

STUDY CHECK

What is the molecular mass of a 4 $g \cdot L^{-1}$ density gas at 30C° and 5 atm.

`STP conditions` STP conditions refer to standard temperature (273K) and pressure (1 atm) conditions. Working at STP conditions means pressure will be fixed at 1 atm and temperature at 273K.

1 atm and 273K STP Conditions

Sample Problem 4

Calculate the volume in litters of 5 moles of nitrogen at STP conditions.

SOLUTION

From the problem we have the following data:

	Given	Asking
Analyze the Problem	$n = 5moles$ $P = 1atm$ $T = 273K$	V

We need to apply the ideal gas formula with the set of given variables:

$$1\cancel{\text{atm}} \cdot V = 5\cancel{\text{mol}} \cdot 0.082 \frac{\cancel{\text{atm}} \cdot \text{L}}{\text{mol} \cdot \cancel{K}} \cdot 273\cancel{K}$$

and solving for V we have a final volume of 112L.

STUDY CHECK

Calculate the grams in 4L of N_2 at STP conditions.

5.3 Change of gas properties

The previous section addressed the properties of an ideal gas. However, as all properties of a gas are related, if we modify one the others will change too. This section covers situations in which one of the gas properties changes (e.g. V changes) and you need to predict the change of another gas property (e.g. P). For example, imagine you compress a balloon with your hand. The temperature and number of moles of the gas inside the balloon are constant, as the balloon is closed and in contact with the atmosphere. Differently, the pressure and volume will change. In particular, the volume will decrease and the pressure will increase. This means that the gas particles will hit the balloon harder and with more frequency.

Solving problems with an initial and final state In order to solve problems in which two of the gas variables are kept fixed and the other two are fixed, one needs to apply the ideal gas law at the initial and final state to then divide both formulas. Imagine the situation in which you have a 1L hot air balloon with 1 moles of a gas and you add gas to a total of 5 moles. You want to calculate the final volume after you inflate the volume, knowing the temperature and pressure are kept constant. The initial state corresponds to 1L and 1 moles of gas and the final estate corresponds to an unknown volume and 5 moles. Using the ideal gas formula twice you have:

$$\left.\begin{aligned} PV_1 &= n_1RT \\ PV_2 &= n_2RT \end{aligned}\right\} \quad \frac{PV_1}{PV_2} = \frac{n_1RT}{n_2RT} \tag{5.1}$$

as some of the variables of the cancel out:

$$\frac{\cancel{P}V_1}{\cancel{P}V_2} = \frac{n_1\cancel{RT}}{n_2\cancel{RT}} \tag{5.2}$$

and you end up with Avogadros' law. If you plug the numbers into the formula:

$$\frac{1L}{V_2} = \frac{1 \text{ mol}}{5 \text{ mol}} \tag{5.3}$$

and you get a final volume of 5L.

Sample Problem 5

A 3L gas sample has a pressure of 5 atm. If the pressure increases to 10 atm at fixed temperature and number of moles, calculate the final volume of the gas.

SOLUTION

From the problem we have the following data:

	Given	Asking
Analyze the Problem	$V_1 = 3L$ $P_1 = 5atm$ $P_2 = 10atm$	V_2

We need to apply the ideal gas formula to the initial state and final state and divide both formulas. The number of moles and the temperature are constant

and will cancel out from both equations:

$$\left.\begin{aligned} P_1V_1 &= nRT \\ P_2V_2 &= nRT \end{aligned}\right\} \quad \frac{P_1V_1}{P_2V_2} = \frac{nRT}{nRT} \tag{5.4}$$

Plugging the values:

$$\frac{P_1V_1}{P_2V_2} = \frac{nRT}{nRT} \tag{5.5}$$

and solving:

$$\frac{3 \cdot 5}{10 \cdot V_2} = 1 \tag{5.6}$$

the final volume will be 1.5 L.

STUDY CHECK

A 4 atm gas sample has a temperature of 300K. If we decrease its temperature to 200K at fixed volume and number of moles, calculate the final pressure of the gas.

`Pressure-Volume change` If temperature and the number of moles of a gas are kept constant the product of pressure and volume will remain constant too. This is the case of the balloon-pressing example. We call this Boyle's Law:

$$\boxed{\frac{P}{V} = c \quad or \quad P_1 \cdot V_1 = P_2 \cdot V_2} \quad \text{Boyle's law}$$

where:

P_1, V_1 are the initial pressure and volume

P_2, V_2 are the final pressure and volume

c is a constant

`Volume-Temperature change` Imagine you cool down a balloon at fixed pressure (under the atmosphere). What would happen to the balloon's volume? Based on Charle's law, its volume will decrease:

$$\boxed{\frac{V}{T} = c \quad or \quad \frac{V_1}{T_1} = \frac{V_2}{T_2}} \quad \text{Charle's law}$$

where:

V_1, T_1 are the initial volume and temperature

V_2, T_2 are the final volume and temperature

c is a constant

`Volume-Moles change` Imagine a hot air balloon. Air comes in and out of the balloon as the balloon is not closed balloon. Hence the pressure inside the balloon is just the atmospheric pressure. Also as the balloon is in contact with the air, its temperature will be constant, resulting from the thermal equilibrium between the inside of the balloon and the atmoshphere. If you inflate the balloon with hot air, the volume of the balloon and the number of moles are related by Avogadro's law:

$$\boxed{\frac{V}{n} = c \quad or \quad \frac{V_1}{n_1} = \frac{V_2}{n_2}} \quad \text{Avogadro's law}$$

where:

V_1, n_1 are the initial volume and number of moles

V_2, n_2 are the final volume and number of moles

c is a constant

`Relating the different variables of a gas` The questions is now, if we increase the pressure at fixed number of moles and pressure, how do we know if the volume will increase or perhaps decrease? Similarly, if for example the number of gas moles increase at fixed pressure and volume, will the temperature of the gas increase or perhaps decrease. We can answer these questions by means of the ideal gas law. If the variables that we need to relate are in the same side of the equation (e.g. P and V) then if one of the variables increase the other will decrease. Differently, If the gas variables to relate are located in opposite sides of the gas law (e.g. P and T) then both will change in the same direction. For example, let us consider the changes of P and V (at fixed n and T). As they are in the same side of the ideal gas law ($PV = nRT$), if P increases V will decrease. Differently, for the change of P and T (at fixed V and n), as both variables are in opposite sides of the ideal gas law ($PV = nRT$), if P increases, T will increase as well.

5.4 Mixtures of gases

The air is a mixture of different gases. It contains oxygen (O_2) and nitrogen (N_2) as well as other gases such as carbon dioxide, argon, or water vapor. Only 21% of the air is made of oxygen and 78.2% of nitrogen. The other gases represent 0.8% of the air. The atmospheric pressure is 1 atm and results from the pressure of all the components of the air. Each gas exerts a partial pressure and all combined exert the total atmospheric pressure. In this section you will learn how to work with mixtures of gases. This section also covers the use of the molar volume to relate moles and volume at standard conditions.

`Partial and total pressure` Imagine you have a container with 1atm of Ar, and nother container of the same volume containing 1 atm of Ne. If you combine the containers into a single container (and temperature does not change), hence the pressure in the container will result from both gases and will be 2 atm. Inside the mixed container, 2 atm will the the total pressure (P_{Total}), whereas the partial pressure of each gas (p_1 and p_2) will be 1 atm. Dalton's Law says that the total pressure results from adding the partial pressure of each gas. For a gas mixture with n components:

$$\boxed{P_{Total} = p_1 + p_2 + \ldots p_n} \quad \text{Dalton's Law}$$

Sample Problem 6

Medical Air is a odorless gas made mostly of nitrogen and oxygen, administer by ventilator in hospital settings with an operating gauge pressure of 3 atm. If the oxygen pressure inside a container is 2.37 atm, calculate the partial pressure of nitrogen in the mixture.

SOLUTION

The problem gives the total pressure of the mixture and the partial pressure of

one of the components. By using Dalton's law, we know that if the total pressure is 3atm and the partial pressure of oxygen is 2.37, hence the partial pressure of the other component has to be 0.63 atm.

STUDY CHECK

Entonox is a medicinal mixture of dinitrogen oxide (N_2O) and oxygen (O_2). The pressure N_2O in a entonox container is 2 atm and the oxygen pressure is 1520 mmHg as well. Calculate the total pressure in atm in a Entonox container.

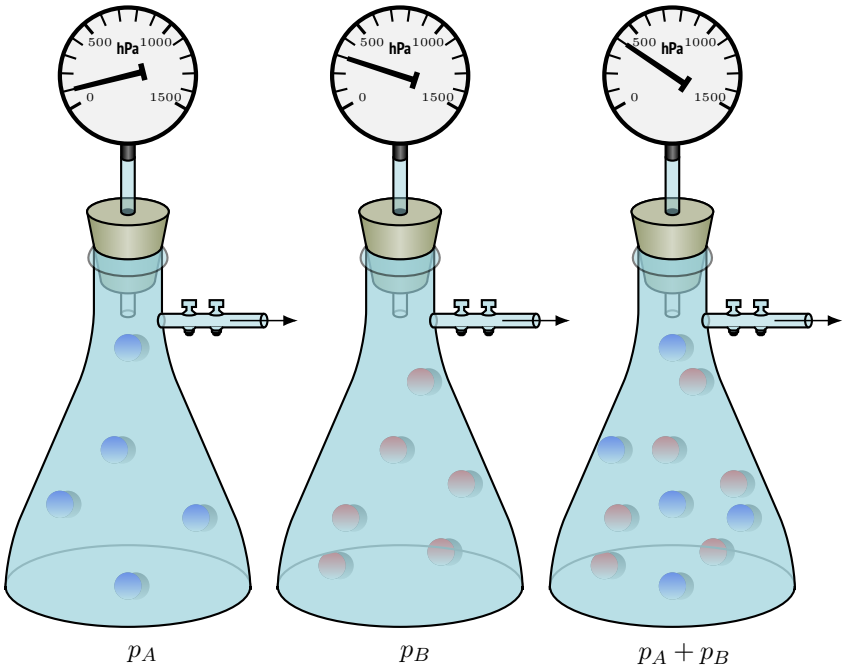


Figure 5.3 A visual representation of Dalton's law of partial pressure: after adding two different gases with different partial pressured, the final pressure is the result of adding both partial pressures.

`Partial pressure of a gas in a mixture` For a mixture with different gases, the partial pressure of a given gas (A) will depend on the number of moles of that particular gas and the overall volume of the mixture

$$\boxed{p_A = \frac{n_A RT}{V}}$$

`Mole fraction` The mole fraction (X_A) of a gas (A) in a mixture of gas is just the number of moles of this gas over the total number of moles in the mixture. The larger the mole fraction of a gas in a mixture the more molecules of that specific gas are there in the mixture with respect to all components. One can express the morel fraction in terms of partial pressures also, as the pressure of a given gas over the total pressure. For a mixture with n components:

$$\boxed{X_A = \frac{n_A}{n_A + n_B + \cdots + n_n} \quad or \quad \frac{p_A}{p_A + p_B + \cdots + p_n}}$$

For a mixture of gases, the partial pressure of a gas (p_A) is related to the mole fraction of that gas (X_A) and the total pressure of the mixture of gases (P_{Total}):

$$\boxed{p_A = X_A \cdot P_{Total}}$$

Sample Problem 7

A mixture of gases with a total pressure of 2 atm contains 3 moles of Ar, 3 moles of He and 1 moles of Ne. Calculate the partial pressure of each component on the mixture.

SOLUTION

We calculate first the mole fraction for each component of the mixture. As the total number of moles if 7 moles and there are 3 moles of Ar, its mole fraction is 0.43. Similarly, the mole fraction for He is 0.43 and for Ne is 0.14. To calculate the partial pressure of each gas you just need to multiply its mole fraction by the total pressure (2 atm). Hence: p_{Ar}=0.86atm, p_{He}=0.86atm and p_{Ne}=0.28atm

STUDY CHECK

A mixture of gases with a total pressure of 5 atm contains 1 mol of Ar and 1 mol of He. Calculate the partial pressure of each component on the mixture.

5.5 Covalent compounds

Covalent compounds are chemicals resulting from the combination of nonmetallic elements. And example is CO_2, which results of combining carbon (a non metal) with oxygen (a non metal).

`Covalent naming` In order to name a covalent compound you need to (a) use the name of the first element in the compound, (b) use the first syllable of the second element, and (c) finish the name of the molecule in the suffix *-ide*. More importantly, you need to use prefixes that indicate the number of atoms in the molecule. See Table **??** for a list of the different equivalencies between prefixes and number. As an example, the formula CH_4 is named as carbon tetrahydride. Similarly, a covalent chemical name can be translated into a formula (we call this to formulate a chemical with a given name), and the formula for carbon monoxide would be CO. When the vowels *a* and *o* appear together, the first vowel is omitted as in carbon monoxide instead of carbon ~~monooxide~~. Another example would be N_2O named as dinitrogen oxide, and the name sulfur hexafluoride corresponds to the formula SF_6. The prefix mono is omitted in the first element of the name, and for example you will not name the chemical CO as ~~monocarbon~~ monoxide, you would just say carbon monoxide. A final example of a covalent compound:

N_2O_5	dinitrogen pentoxide	BCl_3	Boron trichloride

Table 5.1 Prefixes used to name covalent compounds

Prefix	number	Prefix	number
Mono	1	Hexa	6
Di	2	Hepta	7
Tri	3	Octa	8
Tetra	4	Nona	9
Penta	5	Deca	10

Sample Problem 8

Name of give the name of the following covalent chemicals:

Formula	Name
NO	
CS_2	
	Sulfur Dioxide
	Nitrogen Trichloride

SOLUTION

All chemicals in this example are covalent as they result of the combination of nonmetals. In order to name them, we need to use prefixes and finish the sufix with -ide. The first chemical is called nitrogen oxide. CS_2 is called carbon disulfide. The formula for sulfur dioxide and nitrogen trichoride are respectively SO_2 and NCl_3.

STUDY CHECK

Name of give the name of the following covalent chemicals: SCl_2 and diboron thrioxide.

Properties of covalent compounds At normal conditions, covalent compounds may exist as solids, liquids, or gases. Covalent compounds do not exhibit any electrical conductivity, either in pure form or when dissolved in water. A typical covalent compound would be H_2O, water.

The covalent bond Atoms in a covalent compound are connected by means of a covalent chemical bond. In a covalent bond, both atoms connected share the electrons. As an example, the HCl molecule has an hydrogen and a chlorine atom connected by means of a covalent bond, in which each atoms share the electrons of the bond.

5.6 Radioactive gases: radon

Radon is a colorless, odorless radioactive gas produced by the radioactive decay of uranium. It is present in nearly all soils and very small levels of radon are found in the air we breathe every day. The problem occurs when radon gas enters our home and gets trapped. If you are breathing in too much radon, you will not feel sick right away. Only long-term exposure to high levels of radon can cause lung cancer and the risk is higher for those who smoke. While questions still remain over the quantities and length of exposure, radon concerns are a fact of homeownership. Most residential real estate transactions require radon testing, and many states require radon mitigation for new construction. The recommended reference radon level is 100 $Bq \cdot m^3$ in dwellings. Testing your home for radon is easy and doesn't cost very much. You can test for radon yourself or hire a professional to do it for you. There are relatively simple tests for radon gas. Radon detection devices are commercially available. Digital radon detectors provide ongoing measurements giving both daily, weekly, short-term and long-term average readouts via a digital display. Short-term radon test devices used for initial screening purposes are inexpensive, in some cases free.

5.7 Blood as a buffer

We will finally discuss the role of CO_2 as a buffer in the blood.

`Carbon dioxide is an acid` Carbon dioxide is an acid gas. When it dissolved reversibly in water it produces carbonic acid:

$$CO_{2(g)} + H_2O_{(l)} \rightleftharpoons H_2CO_{3(aq)}$$

At the same time H_2CO_3 partially dissociates in water as a weak acid producing bicarbonate–also known as hydrogen carbonate:

$$H_2CO_{3(aq)} + H_2O_{(l)} \rightleftharpoons HCO_3{}^-{}_{(aq)} + H_3O^+_{(aq)}$$

As a result, the dissolution of carbon dioxide in the blood produced a buffer containing carbonic acid and bicarbonate–the conjugate base of carbonic acid. This buffer system keeps the PH of blood constant to a value of 7.4 that is slightly above the neutral value of 7.

`The dangerous change in blood PH` Most biological molecules are very sensitive PH changes. Small deviations in PH can be dangerous or even fatal. If the blood PH falls below 7.34 it becomes more acidic causing a clinical condition called *acidosis*. On the other hand, if the PH rises above 7.45, the blood becomes more basic causing a condition called *alkalosis*. Changed of PH below 6.8 or above 8.0 can cause death.

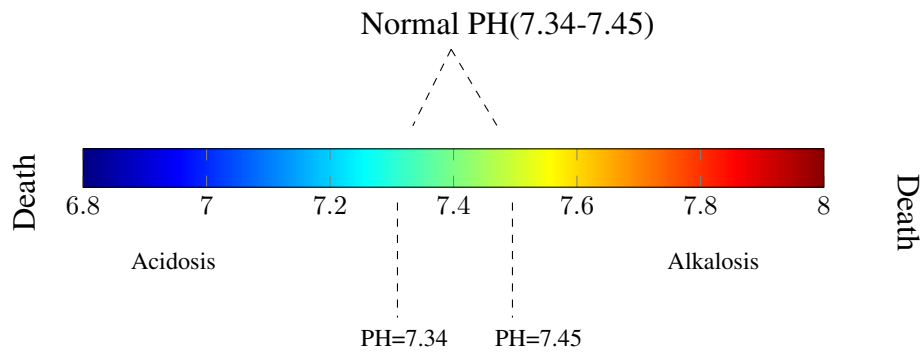


Figure 5.4 PH range that leads to acidosis and alkalosis

`Alkalosis and carbon dioxide` Our body needs oxygen to function properly. When you inhale, you introduce oxygen into the lungs. When you exhale, you release carbon dioxide, which is a waste product. Normally, the respiratory system keeps these two gases in balance. Respiratory alkalosis occurs when you breathe too fast or too deep. This hyperventilation–also known as over-breathing–causes carbon dioxide levels to drop too low. This means there is not enough CO_2 in the blood and that causes the PH to increase and become too alkaline. Treating the condition is a matter of raising carbon dioxide levels in the blood. You can do this by breathing from a plastic bag. Respiratory acidosis occurs when too much CO_2 builds up in the body. Normally, the lungs remove CO2 while you breathe. However, sometimes your body can?t get rid of enough CO_2.Treatments for this condition are usually designed to help your lungs. For example, you may be given drugs to dilate your airway. You might also be given oxygen or a continuous positive airway pressure (CPAP) device. The CPAP device can help you breathe if you have an obstructed airway or muscle weakness.

Sample Problem 9

Explain why CO_2 decreases the PH of blood.

SOLUTION

CO_2 is an acid and when dissolved in water it generates protons, H^+. Increasing the amount of protons decreases PH as PH is a measure the concentration of protons in solution.

STUDY CHECK

The PH of a patients blood is 7.3. Indicate whether the patient suffers alkalosis, acidosis or perhaps none of the affections. Can the PH level cause death?

5.8 Chemical reactions

When we eat we burn food with molecular oxygen (O_2) to produce carbon dioxide and water. Similarly, when we start the engine of the car to go to work, gasoline burns to produce the same chemicals: CO_2 and H_2O. These are two examples of chemical reactions, but there are many other examples. Nitrogen from the air reacts with hydrogen to produce ammonia, a common chemical used in the production of fertilizers. This section covers the basic of chemical reactions. You will learn how to balance reaction and how classify reactions.

Simple chemical reactions Magnesium is a metal that react with oxygen to produce magnesium oxide. Magnesium is solid $Mg_{(s)}$ whereas oxygen is gas and contains two oxygen atoms per molecule $O_{2(g)}$. Magnesium oxide, the result of the reaction, is solid $MgO_{(s)}$. The reaction between magnesium and oxygen to produce magnesium oxide

$$2\,Mg_{(s)} + 1\,O_{2(g)} \longrightarrow 2\,MgO_{(s)}$$

Mg and O_2 combine together–that is why we use a plus sign– to produce MgO–we use an arrow to indicate that a chemical is being produced. Also the symbols *(s)* or *(g)* indicates solid or gas state. The reactants are located before the arrow and the products after the arrow. The numbers in front of the reactants and products (2,1 and 2) are called stoichiometric coefficients, and we will talk more about them in the following sections.

Reading a chemical reaction Chemicals reactions can be read in words. In order to read a chemical reaction you need to connect the reactants with the word "react" and then use the words "to produce" and after that you need to read the products. The numbers in front of the reactants and products represent the number of moles, and you need to include those numbers in the reading. For example, the following reaction

$$2\,Mg(s) + O_2(g) \longrightarrow 2\,MgO(s)$$

should be read as: "two moles of Mg react with one mole of O_2 to produce two moles of MgO".

Balanced chemical reactions Chemical reactions contain molecules, which are made of atoms. Some chemical reactions are balanced, and others need to be balanced. In order to identify a balanced reaction, you should use the stoichiometric coefficients and the indexes in the molecular formulas to break down the

reactants and products into atoms. In a balanced chemical reaction, the atoms of reactants should be the same as the atoms of the products. Consider the following reaction,

$$2\,Mg(s) + O_2(g) \longrightarrow 2\,MgO(s).$$

The table below shows all reactants and products in the form of atoms.

$2\,Mg(s) + O_2(g) \longrightarrow 2\,MgO(s)$		
Reactants	Products	
2Mg	2Mg	✓
O_2=2O	2O	✓

The number of Mg atoms in the reactants and products is the same, and equals to two. On the other hand, the number of O atoms in the reactants and products is the same, being equal to two. For this reason, we say this reaction is *balanced*.
Now consider the following reaction:

$$C(s) + O_2(g) \longrightarrow CO(g),$$

The number of C atoms in the reactants and products is the same, and equals to one. In contrast, the number of O atoms in the reactants and products is not the same, and for this reason, we say this reaction is *not balanced*.

$C(s) + O_2(g) \longrightarrow CO(g)$		
Reactants	Products	
1C	1C	✓
O_2=2O	O	✗

Sample Problem 10

Balance the following reaction:

$$CH_4(g) + O_2(g) \longrightarrow CO_2(g) + H_2O(g)$$

SOLUTION

We will break down each molecule into atoms. In the case of O, both CO_2 and H_2O contain oxygen and hence you will have to combine both oxygen atoms:

$CH_4(g) + O_2(g) \longrightarrow CO_2(g) + H_2O(g)$		
Reactants	Products	
1C	1C	✓
4H	2H	✗
2O	3O	✗

The reaction is not balanced as the number of H and O atoms for the reactants and products is not the same. In order to balance the H, you can multiply by two H_2O, and that will balance H but also affect O.

$CH_4(g) + O_2(g) \longrightarrow CO_2(g) + 2\,H_2O(g)$		
Reactants	Products	
1C	1C	✓
4H	4H	✓
2O	4O	✗

You can balance O by multiplying O_2 by two. That will give you the final

balanced reaction in which all atoms (O, H and C) are the same in the product and reactant molecules.

$$CH_4(g) + 2\,O_2(g) \longrightarrow CO_2(g) + 2\,H_2O(g)$$

Reactants	Products	
1C	1C	✓
4H	4H	✓
4O	4O	✓

STUDY CHECK

Balance the following reaction: $Fe_2O_3(s) + C(s) \longrightarrow Fe(s) + CO(g)$

5.9 Square units and units of volume

Square or cubic units How big is your apartment? You might be living in a $750ft^2$ loft in Brooklyn or in a larger house Upstate. Often times we encounter cubic or square units such as cubic centimeter (cm^3) or square feet (ft^2). The equivalencies for cubic or square units should take into account the unit power (power of two or power of three). If $1cm = 1 \times 10^{-2}m$, for square units the relation should be squared and $1cm^2 = 1 \times (10^{-2})^2m^2 = 1 \times 10^{-4}m^2$. Another example, for the case of mm and mm^3:

$$\boxed{\frac{1mm}{1 \times 10^{-3}m} \quad \text{and} \quad \frac{1mm^3}{1 \times 10^{-9}m^3}}$$

Let us work on an example in which we want to convert $30m^2$ into m^2:

$$30\cancel{m^2} \times \frac{1cm^2}{1 \times 10^{-4}\cancel{m^2}} = 3 \times 10^5 cm^2$$

Sample Problem 11

How many m^2 is $20cm^2$.

SOLUTION

In order to convert $20cm^2$ to square meters, we need to remove the centi prefix and that will give us the unit square meter without any prefix. We will use the conversion factor that relates m^2 to cm^2: $\frac{1 \times 10^{-4}m^2}{1cm^2}$ or $\frac{1cm^2}{1 \times 10^{-4}m^2}$.

$$20\cancel{cm^2} \times \frac{1 \times 10^{-4}m^2}{1\cancel{cm^2}} = 2 \times 10^{-3}m^2$$

STUDY CHECK

Convert $100m^3$ to dm^3.

Units of volume Units such as L or mL are units of volume. As volume is a three-dimensional property, those units somehow have to be related to the units of length. In fact one liter is the same as one dm^3 and one ml is the same as one cm^3. In the allied health field, the units *mL* is also written as *cc* as in cubic centiliters.

$$\boxed{1L = 1dm^3 \text{ and } 1mL = 1cm^3}$$

Let us work on an example in which we want to convert $30cm^3$ into L:

$$30\cancel{cm^3} \times \frac{1\cancel{mL}}{1\cancel{cm^3}} \times \frac{1 \times 10^{-3}L}{1\cancel{mL}} = 3 \times 10^{-2}L$$

Sample Problem 12

Convert 30 m^3 into L.

SOLUTION

In order to convert m^3 into L we just need to remember that the L actually refers to dm^3, therefore is connected to meter. We will first convert m^3 into dm^3 and then dm^3 into L.

$$30\cancel{m^3} \times \frac{1\cancel{dm^3}}{1 \times 10^{-3}\cancel{m^3}} \times \frac{1L}{1\cancel{dm^3}} = 3 \times 10^4 L$$

STUDY CHECK

Convert 40L to cm^3.

CHAPTER 5

GASES AND ITS PROPERTIES

5.1 Convert the following properties: (a) A pressure value of 2 atm into mmHg (b) A pressure value of 3000 Pa into atm (c) A temperature value of 25°C into K

5.2 Convert the following properties: (a) A pressure value of 900 mmHg into torr (b) A temperature value of 400K into °C

IDEAL GAS LAW

5.3 A gas contained in a 3L tank has a pressure of 5 atm at a temperature of 400 K. Calculate the number of moles in the tank.

5.4 Dinitrogen oxide, used in dentistry, is an anesthetic also called laughing gas. What is the pressure in atm of 0.35 moles of N_2O at 22°C in a 5L container?

5.5 A 4 moles sample of gas at 400K has a pressure of 10 atm. Calculate the volume of the sample.

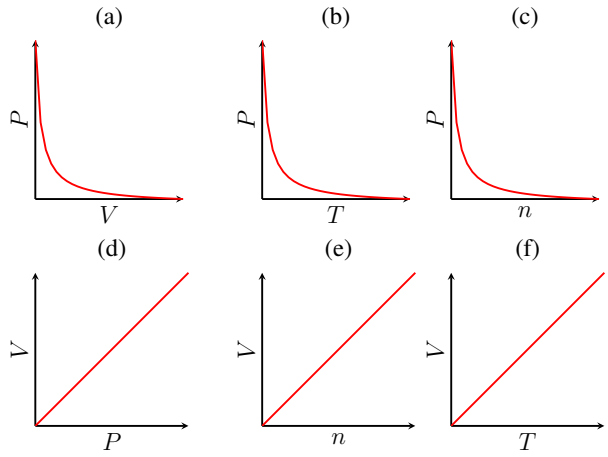
5.6 A 3 grams sample of Ar at 40°C is placed in a 3L container. Calculate the pressure inside the container.

5.7 Eighteen moles of a gas in a 11L container at 400K exert a pressure of 3 atm. Calculate the molar mass of the gas.

5.8 What is the molar mass of a gas if a 3.16 g sample at 0.75 atm and 45°C occupies a volume of 2L.

5.9 Answer the following questions: (a) Calculate the volume of a 4 moles of Ar at STP conditions. (b) Calculate the volume of a 4 moles of Ne at STP conditions. (c) Calculate the moles of gas in 3L of Ar at STP conditions. (d) Calculate the volume of 64 g of O2 gas at STP (273K, 1atm)

5.10 Indicate what plot (or plots) bellow best represent the following gas laws: (a) Boyle's law (b) Charle's law (c) Avogadro's law



CHANGE OF GAS PROPERTIES

5.11 A sample of a gas at 400K and 12 atm is cooled in the same container to 200K. Calculate the new pressure.

5.12 In a storage area where the temperature has reached 300K, the pressure of oxygen gas in a 15 L steel cylinder is 1 atm. Calculate the volume if the pressure is reduced to 0.5 atm.

5.13 A H_2 sample has a volume of 5 L and a pressure of 1 atm. What is the new pressure if the volume is decreased to 2L with no change in temperature and the amount of gas.

5.14 A sample of Ne in a closed, expandable container, has a volume of 3L at 40°C. Calculate the new volume if the container is cooled to 25°C.

5.15 If the pressure of a gas increases, at fixed temperature and moles, its volume....

5.16 If the temperature of a gas increases, at fixed volume and moles, its pressure....

MIXTURES OF GASES

5.17 A tank contains Ne gas at 700 mmHg, Ar at 2 atm, and Kr at 700 torr. What is the total pressure of the mixture in atm?

5.18 An anesthetic consist of a mixture of cyclopropane gas and oxygen gas. If the mixture has a total pressure of 2 atm and the partial pressure of cyclopropane is 0.5atm, what is the partial pressure of O2?

5.19 The atmospheric pressure on a hot day is 780 mmHg. Given that the air is made of 78% of nitrogen and 22% of oxygen, calculate the partial pressure of each gas in the air.

COVALENT COMPOUNDS

5.20 Name or formulate the following covalent compounds: (a) NO (b) Dichlorine monofluoride (c) NO_2

5.21 Name or formulate the following covalent compounds: (a) Chlorine Monofluoride (b) N_2O (c) Nitrogen trifluoride

5.22 Name or formulate the following covalent compounds: (a) SO_3 (b) Disulfur dichloride (c) SO_2 (d) Disulfur tetrachloride

5.23 Name or formulate the following covalent compounds: (a) P_4S_3 (b) Sulfur Tetrafluoride (c) As_2O_5 (d) Sulfur trioxide

CHEMICAL REACTIONS

5.24 Balance the following reactions:

(a) $P_4(s) + O_2(g) \longrightarrow P_4O_{10}(s)$

(b) $Al(s) + O_2(g) \longrightarrow Al_2O_3(s)$

5.25 Balance the following reactions:

(a) $FeS(s) + O_2(g) \longrightarrow Fe_2O_3(s) + SO_2(g)$

(b) $NH_3(g) + O_2(g) \longrightarrow NO(g) + H_2O(g)$

5.26 Classify next reaction as combination, decomposition, single replacement, double replacement, or combustion:

(a) $Pb_{(s)} + FeSO_{4(s)} \longrightarrow PbSO_{4(s)} + Fe_{(s)}$

(b) $C_6H_{12(g)} + 9\,O_{2(g)} \longrightarrow 6\,CO_{2(g)} + 6\,H_2O_{(g)}$

(c) $2\,RbNO_{3(aq)} + BeF_{2(aq)} \longrightarrow Be(NO_3)_{2(aq)} + 2\,RbF_{(aq)}$

5.27 Balance the following reactions:

(a) $H_{2(g)} + Br_{2(g)} \longrightarrow HBr_{(g)}$

(b) $C_{(g)} + O_{2(g)} \longrightarrow CO_{(g)}$

(c) $O_{3(g)} \longrightarrow O_{2(g)}$

(d) $NH_4NO_{2(aq)} \longrightarrow N_{2(g)} + H_2O_{(l)}$

(e) $Na_3PO_{4(aq)} + MgCl_{2(aq)} \longrightarrow Mg_3(PO_4)_{2(aq)} + NaCl_{(aq)}$

SQUARE UNITS AND UNITS OF VOLUME

5.28 Compute the following power of ten calculations:

(a) $(10^2)^2$

(b) $(10^2)^3$

(c) $(10^{-6})^2$

(d) $(10^{-2})^2$

5.29 Set up the conversion factor to convert $400cm^2$ into m^2:

$$400\cancel{cm^2} \times \frac{\square}{\square} = \square\, m^2$$

5.30 Fill the gap in the following unit equalities or conversion factors:

$1cm^2 = \square\, m^2$ $\quad$ $1cm^3 = \square\, m^3$

$1dm^3 = \square\, m^3$ $\quad$ $1dm^2 = \square\, m^2$

Answers **5.1** (a) 1520 mmHg (b) 2.96×10^{-2} atm (c) 298K **5.3** 5 moles **5.5** 1915 L **5.7** 18 g/mol **5.9** (a) 89.6L (b) 89.6L (c) 0.13mol (d) 44.7L **5.11** 6 atm **5.13** 2.5L **5.15** decreases **5.17** 1402torr **5.19** N_2 608 mmHg, O_2 171.6 mmHg **5.21** (a) Chlorine Monofluoride (ClF) (b) N_2O (dinitrogen monoxide) (c) Nitrogen trifluoride(NF_3) **5.23** (a) P_4S_3 (tetraphosporus trisulfide) (b) Sulfur Tetrafluoride (SF4) (c) As_2O_5 (diarsenic pentoxide) (d) Sulfur trioxide (SO3) **5.25** (a) $4\,FeS(s) + 7\,O_2(g) \longrightarrow 2\,Fe_2O_3(s) + 4\,SO_2(g)$ (b) $4\,NH_3(g) + 5\,O_2(g) \longrightarrow 4\,NO(g) + 6\,H_2O(g)$ **5.27** not provided **5.29** $0.04m^2$